

Lecture 7: Remedial measures

Transformations and weighted least square

Phuong L. Vo

International University

September 28, 2026



1 Transformations

2 Weighted least square

- When error variances known
- Error variances unknown

1 Transformations

2 Weighted least square

- When error variances known
- Error variances unknown

Overview of remedial measures

If the simple normal error linear regression model is not appropriate for a data set, there are two basic choices:

- ① Abandon current regression model and develop and use a more appropriate model.
 - entail a more complex model that could yield better insights
 - may also lead to more complex procedures for estimating the parameters
- ② Employ some transformation on the data so that current regression model is appropriate for the transformed data.
 - leads to relatively simple methods of estimation and may involve fewer parameters than a complex model
 - an advantage when the sample size is small.

Nonlinearity of regression function

- When the regression function is not linear, a direct approach is to modify regression model by altering the nature of the regression function.
- For instance, a quadratic regression function might be used:

$$E\{Y\} = \beta_0 + \beta_1 X + \beta_2 X^2$$

or an exponential regression function:

$$E\{Y\} = \beta_0 \beta_1^X$$

- When the nature of the regression function is not known, exploratory analysis that does not require specifying a particular type of function is often useful (Lowess method, subsection 3.10, Kutner's textbook.)

Nonconstancy of error variance

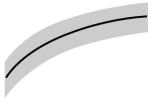
- When the error variance is not constant but varies in a systematic fashion, a direct approach is to modify the model to allow for this and use the method of **weighted least squares** to obtain the estimators of the parameters (Chapter 11, Kutner's textbook).
- Transformations can also be effective in stabilizing the variance, which will be mentioned in next section.

Nonindependence or nonnormality of error terms

- Lack of normality and nonconstant error variances frequently go hand in hand. The same transformation that helps stabilize the variance is also helpful in approximately normalizing the error terms. We discuss transformations to achieve approximate normality in next section.

Transformations for nonlinear relation only

- We consider transformations for linearizing a nonlinear regression relation.
- Apply when the distribution of the error terms is reasonably close to a normal distribution and the error terms have approximately constant variance

	Prototype Regression Pattern	Transformations of X
(a)		$X' = \log_{10} X$ $X' = \sqrt{X}$
(b)		$X' = X^2$ $X' = \exp(X)$
(c)		$X' = 1/X$ $X' = \exp(-X)$

Transformations for nonlinear relation only - example

An experiment on the effect of number of days of training received (X) on performance (Y) in a battery of simulated sales situations for the 10 participants.

Sales Trainee	(1) Days of Training	(2) Performance Score	(3)
i	X_i	Y_i	$X'_i = \sqrt{X_i}$
1	.5	42.5	.70711
2	.5	50.6	.70711
3	1.0	68.5	1.00000
4	1.0	80.7	1.00000
5	1.5	89.0	1.22474
6	1.5	99.6	1.22474
7	2.0	105.3	1.41421
8	2.0	111.8	1.41421
9	2.5	112.3	1.58114
10	2.5	125.7	1.58114

- We apply the transformation $X' = \sqrt{X}$ (Fig.a)

Transformations for nonnormality and unequal error variances

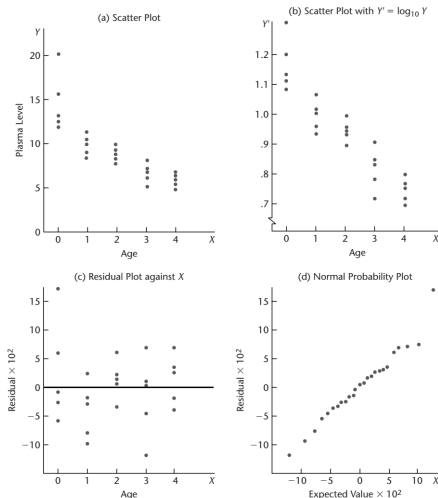
- Unequal error variances and nonnormality of the error terms frequently appear together.
- Frequently, the nonnormality and unequal variances cause an increasing skewness and increasing variability of the distributions of the error terms as the mean response $E\{Y\}$ increases.
- To remedy these departures from the simple linear regression model, we need a transformation on Y , since the shapes and spreads of the distributions of Y need to be changed.

Transformations for nonnormality and unequal error variances - Example

Child i	(1) Age X_i	(2) Plasma Level Y_i	(3) $Y'_i = \log_{10} Y_i$
1	0 (newborn)	13.44	1.1284
2	0 (newborn)	12.84	1.1086
3	0 (newborn)	11.91	1.0759
4	0 (newborn)	20.09	1.3030
5	0 (newborn)	15.60	1.1931
6	1.0	10.11	1.0048
7	1.0	11.38	1.0561
...	...	...	...
19	3.0	6.90	.8388
20	3.0	6.77	.8306
21	4.0	4.86	.6866
22	4.0	5.10	.7076
23	4.0	5.67	.7536
24	4.0	5.75	.7597
25	4.0	6.23	.7945

- Data on age (X) and plasma level of a polyamine (Y) for a portion of the 25 healthy children in a study are presented in columns 1 and 2
- Use of logarithmic transformation of Y to linearize regression relation and stabilize error variance - Plasma levels example.

Scatter plots and residual plots - Plasma levels example



Box-Cox transformations

- It is often difficult to determine from diagnostic plots, e.g., in Plasma levels example.
- The Box-Cox procedure automatically identifies a transformation from the family of power transformations on Y

$$Y' = Y^\alpha,$$

where α is a parameter to be determined from the data.

- The model becomes:

$$Y^\alpha = \beta_0 + \beta_1 X_i + \epsilon_i.$$

- The Box-Cox procedure uses the method of maximum likelihood to estimate α , as well as the other parameters β_0 , β_1 , and σ^2 .

Reading: <https://online.stat.psu.edu/stat462/node/85/>

1 Transformations

2 Weighted least square

- When error variances known
- Error variances unknown

Generalized multiple regression model

- The generalized multiple regression model is

$$Y_i = \beta_0 + \beta_1 X_{i1} + \dots + \beta_{p-1} X_{i,p-1} + \epsilon_i$$

where

- $\beta_0, \beta_1, \dots, \beta_{p-1}$ are parameters
- $X_{i1}, \dots, X_{i,p-1}$ are known constants
- ϵ_i are independent $N(0, \sigma_i^2)$ and the variance-covariance matrix is

$$\text{var}\{\epsilon\} = \begin{bmatrix} \sigma_1^2 & 0 & \dots & 0 \\ 0 & \sigma_2^2 & \dots & 0 \\ 0 & 0 & \dots & \sigma_n^2 \end{bmatrix}$$

- The different Y observations for the n cases no longer have the same reliability (variances).

1 Transformations

2 Weighted least square

- When error variances known
- Error variances unknown

Weighted least squares

- We first consider the estimation of the regression coefficients when the *error variances* σ^2 are known. This case is usually unrealistic, but it *provides guidance as to how to proceed when the error variances are not known*.
- Minimizing *weighted least squares*

$$Q_w = \sum_{i=1}^n w_i (Y_i - \beta_0 - \beta_1 X_{i1} - \dots - \beta_{p-1} X_{i,p-1})^2,$$

where $w_i = \frac{1}{\sigma_i^2}$.

- An observation Y_i that has a large variance receives less weight than another observation that has a smaller variance.

Matrix form of weighted least square

- Define the matrix:

$$\mathbf{W} = \begin{bmatrix} w_1 & 0 & \dots & 0 \\ 0 & w_2 & \dots & 0 \\ 0 & 0 & \dots & w_n \end{bmatrix}$$

- The regression equation is $(\mathbf{X}'\mathbf{W}\mathbf{X})\mathbf{b}_w = (\mathbf{X}'\mathbf{W}\mathbf{Y})$
- Therefore, the vector of the estimated regression coefficients is:

$$\mathbf{b}_w = (\mathbf{X}'\mathbf{W}\mathbf{X})^{-1}(\mathbf{X}'\mathbf{W}\mathbf{Y})$$

- The variance-covariance matrix of the coefficients is:

$$\text{var}\{\mathbf{b}_w\} = (\mathbf{X}'\mathbf{W}\mathbf{X})^{-1}$$

- Note that all these are known since the variances σ_i^2 are assumed to be known.

1 Transformations

2 Weighted least square

- When error variances known
- Error variances unknown

When error variances is unknown

- We estimate the variance or the standard deviation from the residuals.
- We fit the regression model using unweighted least squares and then regress the squared or absolute residuals e_i^2 against X .

Use of estimated variance or standard deviation

- After the variance function or the standard deviation function is estimated, the fitted values from this function are used to obtain the estimated weights:
 - $w_i = \frac{1}{(\hat{s}_i)^2}$, where $\hat{s}_i$ is fitted value from standard deviation function
 - $w_i = \frac{1}{\hat{v}_i}$, where $\hat{v}_i$ is fitted value from variance function
- The estimated weights are then placed in the weight matrix **W** and the estimated regression coefficients are obtained by

$$\mathbf{b}_w = (\mathbf{X}'\mathbf{W}\mathbf{X})^{-1}(\mathbf{X}'\mathbf{W}\mathbf{Y})$$

Use of estimated variance or standard deviation: Estimation process

We summarize the estimation process:

- 1 Fit the regression model by unweighted least squares and analyze the residuals.
- 2 Estimate the variance function or the standard deviation function by regressing either the squared residuals or the absolute residuals on the appropriate predictor(s).
- 3 Use the fitted values from the estimated variance or standard deviation function to obtain the weights w_i .
- 4 Estimate the regression coefficients using these weights.

Iteratively reweighted least squares

- If the estimated coefficients differ substantially from the estimated regression coefficients obtained by ordinary (unweighted) least squares, we can iterate the weighted least squares process by using the residuals from the weighted least squares fit to re-estimate the variance or standard deviation function to obtain revised weights
- Often one or two iterations are sufficient to stabilize the estimated regression coefficients.

See:

- <https://online.stat.psu.edu/stat501/lesson/13/13.1>
- <https://www.statsmodels.org/v0.12.2/examples/notebooks/generated/wls.html>
- https://www.statsmodels.org/dev/generated/statsmodels.regression.linear_model.WLS.html

Summary

- Transformation
- Weight least square

References

- Chapters 8, 11, Kutner textbook
- Chapter 5, Montgomery textbook

Thank you.